

EE/RPF

For use from May/November 2027

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International Baccalaureate®
Baccalauréat International
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Student personal code:

Extended essay - Reflection and progress form

Student: This form is to be completed by you and your supervisor during the progress and completion of the extended essay. It records your progress through the extended essay process and includes your reflective statement addressing your extended essay experience. You must undertake three formal reflection sessions with your supervisor: The first formal reflection session should focus on your initial ideas and how you plan to undertake your research; the interim reflection session takes place once a significant amount of your research has been completed, and the final session will be in the form of a viva voce, once you have completed and handed in your extended essay. This document acts as a record in supporting the authenticity of your work. Following the viva voce, you write a **reflective statement** of no more than 500 words., which must be included on the RPF.

The completion of this form is a mandatory requirement of the extended essay. It must be submitted together with the completed essay. The reflective statement will then be assessed under Criterion E. Students and supervisors should be aware that a mark of 0 will be awarded by the examiner for criterion E if the RPF is blank or the reflective statement is written in a language other than that of the accompanying essay.

Supervisor: You must have three reflection sessions with each student, one early in the process, an interim meeting and then the final viva voce. Other check-in sessions are permitted but do not need to be recorded on this sheet. After each reflection session you must record the month that the reflection session took place, the student's year of DP study at that time and initial the form.

By submitting this student work for assessment, you are taking responsibility for its authenticity. No piece of student work should be uploaded/submitted to the e-Coursework system if its authenticity is in doubt.

First reflection session

Month: DP year (1 or 2): Supervisor initials:

Interim reflection session

Month: DP year (1 or 2): Supervisor initials:

Final reflection session – Viva voce

Month: DP year (1 or 2): Supervisor initials:

Reflective statement

(No more than 500 words, written in the same language as the accompanying extended essay)

Human physiology is a topic that really interests me, so I promptly chose biology. As I wanted my research to be inseparable from our daily lives. I thought of food and decided to research the effect of food additives on the human digestive system. Since I decided to do primary research, I needed to set up a methodology to make sure that it is achievable. One huge concern was how to simulate the digestive system in a feasible way, which was difficult as most investigations are in vivo studies on animals and humans. I investigated various academic journals to find out how I could develop an in vitro methodology. However, a challenge was how most in vitro models, which I initially thought were doable, are infeasible both in terms of time and resources. The difficulty of the experiments from academic journals made me realize that my scope of investigation was too broad. Afterwards, I found out the importance of gut microbiota in gut health and body health, so I narrowed down my topic to the effects of food additives on gut microbiota. While reading different sources, I came across an interesting piece of literature investigating synergism. Thus, I decided to research the synergistic effects of the combinations of food additives to add another layer of investigation. After designing my experimental methodology, I conducted pilot testing, and the methodology was quite reliable as visible results were seen. I would continue to enhance my methodology to produce optimal results.

I am very pleased with how my essay turned out. Trying to strike a balance between the significance of my research and my ability to undertake it, I learnt how to narrow down my area of research into proxies that are focused enough to formulate a research question, like using *Lactobacillus acidophilus* as a proxy to represent gut microbiota. The biggest challenge was developing my own methodology, through which I discovered how to explore individual components and then use all information holistically. I learnt how actively initiating communication and time management are critical when writing an essay. Also, there was an anomaly as the results of MSG did not align with my original expectations supported by academic journals. Thus, I realized the issue of motivated perception. Our tendency to find evidence for what we are predisposed to believe in may impact how holistically we gather information. Furthermore, I exceeded my initial plan to explore synergism. Without much research in this area, I took the risk and managed to obtain results that drew conclusions beyond my hypothesis. I also extended my knowledge as I discovered the profound impact of the microbiota and potential consequences of the ubiquitous food additives, all of which I will be more aware of from now on.